

Ankit Kumar

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Backend and systems engineer specializing in concurrency, performance, and real-time infrastructure. Built a Redis-compatible engine (81K ops/sec) and a caching system sustaining 50K RPS; focused on bottleneck analysis and production-grade correctness.

TECHNICAL SKILLS

Languages Java, C++, Rust, SQL
Backend Spring Boot, Java NIO, Virtual Threads (Java 21), REST APIs
Systems Concurrency, Event-driven I/O, WebSocket, WebRTC, Replication Protocols
Databases PostgreSQL, Redis
Infra Prometheus, Grafana, AWS, Oracle Cloud Infrastructure, Git

ACHIEVEMENTS

- **TCS CodeVita Season 13 — Global Rank 15** out of 669,000+ participants
- **Codeforces Expert** (rating 1655) · **LeetCode** — active problem solver across DSA domains

WORK EXPERIENCE

Software Engineer Intern

Feb 2025 – May 2025

ControlShift Talent Pvt. Ltd.

- Architected real-time infrastructure for a live **AI interviewer**: built WebRTC signaling server and WebSocket layer with heartbeat, room management, and reconnection for candidate-AI video sessions
- Offloaded blocking email I/O via job queues with retry and dead-letter handling; built 30+ REST endpoints with centralized error handling and 25+ unit tests on critical paths

PROJECTS

Axiom — Redis-Compatible In-Memory Store

github.com/ankitkr-04/axiom

Java, NIO, Concurrency

- Delivered **~81K GET ops/sec** and **~68K SET ops/sec** (50 concurrent clients; $p99 \leq 3.1\text{ms}$) via a single-threaded NIO event loop with direct ByteBuffer access, eliminating thread contention and GC pressure
- Implemented 29+ Redis-compatible commands: Strings, Lists, Sorted Sets (skip-list), Geospatial (geohash), Streams, and Pub/Sub with pattern matching
- Replicated Redis PSYNC protocol: RDB snapshot on initial sync, async incremental streaming, and WAIT-based sync guarantees; tracked per-replica lag via REPLCONF GETACK

Hyperlinkr — High-Performance URL Shortener

github.com/ankitkr-04/hyperlinkr

Rust, Axum, DragonflyDB

- Designed a low-latency URL shortener with a 4-layer cache (NUMA-aware L1/L2 with TinyLFU, Bloom filter, DragonflyDB), achieving **~50K RPS** on an i3/8GB machine with >95% cache hit rate
- Identified BASE62 code generation as the system bottleneck (458K ops/sec single-threaded); diagnosed Bloom filter atomic contention under concurrency via flamegraph profiling
- Instrumented IP-based rate limiting, real-time analytics pipeline, and circuit-breaker failover on the DragonflyDB persistence layer

TicketLedger — High-Concurrency Cinema Booking Platform

github.com/ankitkr-04/ticket-ledger

Java 21, Spring Boot, PostgreSQL, Stripe

- Enforced zero double-bookings via SELECT FOR UPDATE with strict lock ordering; analytically bounded hot-seat throughput at **~200 ops/sec** per row from lock-commit latency
- Scaled API layer with Java 21 Virtual Threads: 1M parked connections per JVM at ~1 KB/thread, isolating HikariCP (20 connections) as the sole bottleneck
- Guaranteed exactly-once payment execution via idempotent Stripe webhook processing with strict financial consistency over UX in late-payment edge cases

EDUCATION

B.Tech in Computer Science

Aug 2023 – May 2026

Lakshmi Narain College of Technology, Bhopal

CGPA: **8.38 / 10**

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Developer Professional (1Z0-1084-25) — Oracle